A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 35

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1 A:** Its total annual discharge | 1 |
| **01.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Public supplies cannot match consumer demand / the water is often cleaner than rivers<br>**1** It also reduces urban flooding after heavy rain<br>– surplus can be used for aquifer recharge<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquifer** → a permeable rock that stores groundwater / underground water store | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** The reservoir is created in the estuary so no farmland / settlements need to be flooded<br>**1** But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5. | |
| **02.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1** C<br>**Also accept**<br>• **Critical Group** → the people most at risk / most heavily exposed | 1 |
| **02.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate<br>**1** No need to sample sites that were never likely to be polluted<br>– sampling also checks the accuracy of the predictions<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **accuracy** → close to the true value | 2 |
| **02.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify<br>**1** So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **persistent** → does not break down / stays in the environment for a long time<br>**Do not credit**<br>• the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg $dm^{-3}$ at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg $dm^{-3}$ at 4 and 28 h)<br>**1** The night-time minimum is lower on the second night (2.5 vs 3.5)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **03.2** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom)<br>**1** During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level)<br>**1** At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen<br>**1** Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen<br>– warm summer water holds less oxygen and decay is rapid<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **eutrophication** → nutrient enrichment of water leading to algal growth<br>**Do not credit**<br>• 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae | 4 |
| **03.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate<br>**1** At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **deoxygenation** → loss of dissolved oxygen from the water | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.4** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** Cyanobacteria release toxins harmful to livestock, pets and people<br>**1** Floating algae shade submerged / rooted macrophytes which die, disrupting food webs<br>**1** Algae are an unreliable food source (die off rapidly) so food webs are poor<br>**1** Blocked filters / unpleasant taste and smell in water supplies<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9. | |
| **04.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned<br>**1** HCFCs to be phased out (by 2030)<br>– some essential uses (eg halon fire extinguishers in aircraft) still permitted<br>**1** A fund to help (developing) countries implement the Protocol<br>– legally binding targets (unlike the 1985 Vienna Convention)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **CFCs** → CFCs / halogenated organic compounds releasing chlorine<br>**Do not credit**<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 3 |
| **04.2** | **2 marks per point**<br>2 marks for each of the 2 points below; award 1 mark where a point is only partly made.<br>**2 Aerosols:** pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk<br>**2 Refrigerants:** HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** CFCs are drained from the appliance and incinerated / broken down at high temperature<br>**1** Products are CO2 and acidic gases (HCl, HF) which are neutralised with crushed lime<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **CFCs** → CFCs / halogenated organic compounds releasing chlorine<br>**Do not credit**<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 2 |
| **04.4** | 1 mark for the point below.<br>**1** CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years<br>– some ODSs (eg dichloromethane) are not banned<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **ozone layer** → the ozone in the stratosphere that absorbs UV<br>• **CFCs** → CFCs / halogenated organic compounds releasing chlorine<br>• **persistent** → does not break down / stays in the environment for a long time<br>**Do not credit**<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3. | |
| **05.1** | **2 marks per point**<br>2 marks for each of the 3 points below; award 1 mark where a point is only partly made.<br>**2 Ploughing:** increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster<br>– carbon in the soil DOM store falls and atmospheric CO2 rises<br>**2 Draining:** waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly<br>– drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO2 (though methane release stops)<br>**2 Marine pollution:** kills phytoplankton so less dissolved CO2 is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton<br>– less CO2 then dissolves from the atmosphere to replace it, so atmospheric CO2 rises<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **waterlogged** → soil pores filled with water so there is no air | 6 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.2** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1** Rice paddy fields<br>**1** Landfill sites<br>**1** Anaerobic sediments in reservoirs<br>**1** Livestock intestines (cattle)<br>**1** Manure / slurry stores<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas | 3 |
| **05.3** | 1 mark for the point below.<br>**1** More $CO_2$ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions<br>– more $H^+$ lowers the pH<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2. | |
| **06.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** Total = 42 + 26 + 31 + 9 + 22 = 130 kW<br>**1** 31 / 130 x 100 = 23.8% (accept 24%)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (%) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1 Cavity wall insulation:** the foam or fibre traps air in small pockets so convection in the cavity is prevented and the low thermal conductivity of still air reduces conduction through the wall<br>**1 Low-emissivity double glazing:** the still air or inert gas between the panes conducts heat poorly, and the low-emissivity coating reflects infrared radiation back into the room instead of letting it radiate out<br>**1 Heat exchanger:** outgoing stale warm air passes next to, but does not mix with, incoming cold fresh air, so most of the heat is transferred to the incoming air and the building can be ventilated without losing that heat<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infrared** → heat radiation / long-wave radiation re-emitted by the Earth | 3 |
| **06.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Each extra layer saves less heat than the one before, because the layers already in place carry most of the temperature difference, so the saving per unit thickness falls<br>**1** Beyond a certain thickness the cost and the embodied energy of making and fitting more insulation are never repaid by the energy it saves<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **embodied energy** → all the energy used to make and transport a product | 2 |
| **06.4** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** A large power station runs at higher temperatures and pressures and uses large efficient turbines, so it converts a greater proportion of the fuel energy into electricity<br>**1** The heat rejected by a large station is low-grade and the station is usually too far from any user, so it is lost to a river or cooling tower<br>**1** A CHP unit is in the building that needs the heat, so the heat rejected by its generator is piped straight to the heating and hot water, and 70 - 80% of the fuel energy is used rather than about 35%<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **combined heat and power** → using the waste heat from generation for heating | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.5** | **Any 5 of 6**<br>1 mark each for any 5 of the 6 points below, in any order, maximum 5.<br>**1** Occupancy sensors and programmable thermostats so that lighting and heating run only where and when people are present<br>**1** LED lighting, which converts a far greater proportion of the electricity into light and much less into waste heat than filament lamps<br>**1** Large south-facing windows and a thermally massive structure to make use of passive solar gain, with shading to prevent overheating in summer<br>**1** Heating water on demand at the point of use, so no energy is wasted keeping a tank hot and none is lost in the pipe run<br>**1** Choosing materials with low embodied energy, and re-using the existing structure rather than demolishing and rebuilding<br>**1 Changes in behaviour:** turning equipment off rather than leaving it on standby, closing doors, and setting the thermostat a degree lower<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **embodied energy** → all the energy used to make and transport a product<br>**Do not credit**<br>• Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. | 5 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-235; AQA spec 3.3.4.2.2-3.3.4.2.3. | |
| **07.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1 C:** Organic (methyl) mercury<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 1 |
| **07.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury<br>**1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Most heavy metals are less soluble at high pH<br>**1** So they remain immobile and are not leached into water or absorbed by organisms<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **leached** → nutrients or ions washed down out of the soil / washed away by water<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11. | |
| **08.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** t-test<br>**1** The data are measured (continuous) values and two means are being compared<br>**1** The data are (assumed to be) normally distributed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **t-test** → a test for a difference between the means of two normally distributed sets<br>**Do not credit**<br>• 'the result was significant' without comparing the calculated value with the critical value at p = 0.05 | 3 |
| **08.2** | 1 mark for the point below.<br>**1** Use the Mann-Whitney U test instead (compares medians)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data<br>**Do not credit**<br>• 'the result was significant' without comparing the calculated value with the critical value at p = 0.05 | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **08.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** There is a 2% probability that the difference occurred by random chance<br>**1** As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **random** → positions chosen using random numbers so the choice is not biased<br>• **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance<br>• **null hypothesis** → a statement that there is no significant difference or correlation<br>**Do not credit**<br>• 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it | 2 |
| **08.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4<br>**1** About 68% of values lie within one SD of each mean so the two data sets are clearly separated<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **08.5** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists<br>**1** Another variable (eg topography / aspect of the sites) could cause both – a controlled experiment or further evidence is needed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance<br>• **causal** → one variable actually causing the change in the other | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **09.1** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** They store chemical energy that is easy to keep until it is needed, and easy to convert into the heat that is usually wanted<br>**1** They have a high energy density, so they reach the high temperatures needed to smelt metal ores and to raise high-pressure steam for turbines<br>**1** High energy density also makes them efficient to transport, because a small mass or volume carries a lot of energy<br>– 75 t of aviation fuel can carry a 400 t aircraft 5600 km<br>**1** They were often found in very abundant local deposits, so industry could grow up on the coalfield that supplied it<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |
| **09.2** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** 75 t / 400 passengers = 0.1875 t per passenger<br>**1** = 188 kg (accept 187 - 190 kg)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (kg) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 2 |
| **09.3** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Many deposits are too deep, too small or in places that are too difficult to reach for the technology available<br>**1** A lot of natural gas is held in fine-grained impermeable shale, from which it cannot flow to a conventional well<br>**1** Oil shale holds more than the world's total reserves of crude oil, but the hydrocarbons are solid and insoluble, so the rock has to be heated to melt the oil before it will flow<br>**1** Reserves count only what can be extracted economically, so deposits that would cost more than the fuel is worth are excluded<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **impermeable** → does not let water through | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **09.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Economic decisions are made on the cheapest option available at the time rather than on long-term supply, so the mines closed while coal was still in the ground<br>**1** The abandoned mines have flooded and are unsafe, so the remaining coal cannot be reached even though North Sea gas will be exhausted within the next thirty or forty years<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **09.5** | **Any 5 of 7**<br>1 mark each for any 5 of the 7 points below, in any order, maximum 5.<br>**1** Deep mining damages the surface through loss of habitat, the transport infrastructure it needs and subsidence above the worked seams<br>**1** Opencast mining causes greater habitat damage because all the overburden above the coal has to be cleared<br>**1** Spoil heaps from mining occupy land, can be unstable and are slow to revegetate<br>**1** Oil spills from rigs pollute the sea, and oil-based drilling mud used to lubricate the drill pipes can pollute groundwater, rivers and the sea<br>**1** Surplus gas on oil rigs is flared to reduce the risk of explosion, releasing sulfur dioxide, carbon dioxide and smoke<br>**1** Pipeline construction damages habitat over a long, wide corridor, although careful route planning, soil replacement and replanting reduce this<br>**1** Fracking to release shale gas uses large volumes of water, risks contaminating groundwater and can cause local earth tremors<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Fracking** → pumping fluid into rock at high pressure to release gas or oil | 5 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-190; AQA spec 3.3.4.1. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.1** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Principles:** dB is logarithmic (10 dB = 10x; 3 dB = 2x) so small reductions in dB are large reductions in intensity<br>– effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans)<br>• **Aircraft design:** high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing<br>• **Aircraft operation:** take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count<br>• **Airport design / location:** away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation<br>• **Railway:** track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes<br>• **Road:** porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing<br>• **Industrial:** silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection<br>• **Domestic:** acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders<br>• **Evaluation:** which approaches are most effective / cost-effective<br>– conflicts (airports near cities for convenience)<br>– legislation and monitoring (Leq 57 dB contours) needed to enforce<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content. | 9 |
| | • Any appropriate named example, including ones that are not listed here.<br>• **quota** → a legal limit on the mass or number that may be caught<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | |
| | **Model answer - Level 3 (9 marks)** | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | Because the decibel scale is logarithmic, a 10 dB cut is a tenfold reduction in intensity and 3 dB halves it, so modest-sounding design changes matter. Aircraft engines have been made quieter by high-bypass-ratio turbofans, chevron nozzles, hush kits and acoustic liners, with fairings and composite airframes reducing airframe noise. Operation can be changed too: steeper take-off angles and flight paths that avoid housing, constant-descent approaches, restricted reverse thrust, night curfews, Chapter noise limits enforced by fines and quota counts. Airport location and layout - away from population centres, engine-test areas and taxiways kept from housing, acoustic walls and baffle mounds, land-use planning and free insulation for residents - deal with the noise that remains. | |
| | Railways use track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication on curves and composite brakes. Roads use porous or rubber-crumb asphalt, acoustic fences and embankments, tree planting, engine insulation and aerodynamics, traffic routing and double glazing. Industry fits silencers to compressors, presses rather than stamps, uses nylon rollers, drills instead of pile-driving, surrounds piling with bubble curtains to protect cetaceans, times blasting, hangs acoustic mats and operates remotely, with ear protection as a last line. Domestic appliances use acoustic absorbers and volume limiters. | |
| | Design changes at source are the most effective because they reduce noise for everyone; operational limits depend on enforcement; location works only for new developments, and airports are still sited near cities for convenience. Legislation and monitoring of noise contours (57 dB $L_{eq}$) are what turn these options into actual reductions. | |
| | **Weaker answer - Level 2 (5 marks)** | |
| | Noise pollution can be reduced in different ways. Aircraft can be designed with quieter engines and can fly at night less often. Airports can be built away from cities. Roads can have sound barriers next to them and quieter road surfaces. Trains can have their tracks polished and use better brakes. Factories can use quieter machines and workers can wear ear defenders. Noise can also be reduced by planting trees. These methods reduce the noise that reaches people. | |
| | **What makes the difference** | |
| | • Level 2 gives correct but generic examples ('quieter engines'); Level 3 names the design features (high bypass ratio, chevron nozzles, acoustic liners) the question's 'named examples' requires. | |
| | • Level 3 organises by design / operation / location as the question asks; Level 2 mixes them. | |
| | • The decibel-scale point and the closing evaluation of which approaches work best lift Level 3 into the top band. | |
| **10.2** | 1 mark for the point below.<br>**1** The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13. | |
| **11.1** | **Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration<br>– hydrological cycle evaporation-precipitation<br>• **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds<br>• **Tipping points:** self-sustaining processes<br>– why control must precede them<br>– examples<br>• **Implications:** urgency and precautionary principle<br>– need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation<br>– geoengineering risks<br>• Difficulty of prediction because feedbacks interact at different scales<br>– models<br>– international agreements and per capita footprints<br>• Judgement on whether current strategies are adequate | 25 |
| | *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.2 | **Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Composition and structure:** $N_2$, $O_2$, Ar, $CO_2$, ozone<br>– troposphere and stratosphere<br>– pressure and liquid water<br>– gases for biological molecules and industry<br>• **Energy processes:** UV absorbed by O / $O_2$ / $O_3$ dynamic equilibrium<br>– visible light absorbed at the surface<br>– infrared absorbed by greenhouse gases (33 °C warming)<br>– heat distribution by winds and ocean currents<br>– transport of water vapour<br>– energy budget and albedo<br>• **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species<br>– difficulties of monitoring and prediction<br>– control strategies and agreements<br>• **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time<br>• **Other threats:** aerosols, smoke, land-use change<br>– interconnection of processes | 25 |
| | • **Evaluation:** which functions are most at risk<br>– success of ozone controls vs slow progress on greenhouse gases<br>– justified conclusion<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1. | |

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